

Canonical Cover and Extraneous Attributes in Relational Databases

1 Extraneous Attributes

1.1 What Are Extraneous Attributes?

Definition

Given a set F of functional dependencies and a dependency $\alpha \rightarrow \beta$ in F :

- An attribute A is **extraneous in the left side** α if $A \in \alpha$ and

$$F \models (F - \{\alpha \rightarrow \beta\}) \cup \{(\alpha - A) \rightarrow \beta\}$$

That means we can remove A from the left side without changing the meaning.

- An attribute A is **extraneous in the right side** β if $A \in \beta$ and

$$(F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - A)\} \models F$$

That means we can remove A from the right side without changing the meaning.

Note: The implication in the opposite direction is trivial because stronger dependencies always imply weaker ones.

1.2 Examples

Examples of Extraneous Attributes

- Given

$$F = \{A \rightarrow C, \quad AB \rightarrow C\}$$

B is extraneous in $AB \rightarrow C$ because $A \rightarrow C$ already holds and implies $AB \rightarrow C$.

- Given

$$F = \{A \rightarrow C, \quad AB \rightarrow CD\}$$

C is extraneous in the right side of $AB \rightarrow CD$ because $AB \rightarrow C$ still holds after removing D .

1.3 How to Test if an Attribute is Extraneous

1. To test if $A \in \alpha$ is extraneous in the **left side** α :
 - Compute $(\alpha - A)^+$ using dependencies in F .
 - If $(\alpha - A)^+$ contains β , then A is extraneous in α .
2. To test if $A \in \beta$ is extraneous in the **right side** β :
 - Let $F' = (F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - A)\}$.
 - Compute α^+ using only dependencies in F' .
 - If α^+ contains A , then A is extraneous in β .

Example: Testing Extraneous Attributes

Consider the set of functional dependencies:

$$F = \{A \rightarrow BC, \quad B \rightarrow C, \quad AB \rightarrow C\}$$

Testing if A is extraneous in the left side of $AB \rightarrow C$:

- Compute closure of $(AB - A) = B$, i.e. compute B^+ :

$$B^+ = \{B\}$$

Using $B \rightarrow C$, add C :

$$B^+ = \{B, C\}$$

- Since B^+ contains C , it means that B alone can determine C .
- Therefore, A is **extraneous** in the left side of $AB \rightarrow C$.

Testing if C is extraneous in the right side of $A \rightarrow BC$:

- Define:

$$F' = (F - \{A \rightarrow BC\}) \cup \{A \rightarrow B\} = \{A \rightarrow B, \quad B \rightarrow C, \quad AB \rightarrow C\}$$

- Compute A^+ using F' :

$$A^+ = \{A\}$$

Using $A \rightarrow B$:

$$A^+ = \{A, B\}$$

Using $B \rightarrow C$:

$$A^+ = \{A, B, C\}$$

- Since A^+ in F' contains C , C is **extraneous** in the right side of $A \rightarrow BC$.

2 Canonical Cover

2.1 What is Canonical Cover?

Definition

Sometimes, a set of functional dependencies (FDs) contains redundant information — some dependencies or parts of them can be removed without changing the meaning.

A **canonical cover** is a simplified, minimal set of functional dependencies that represents the same rules as the original set, but without any redundancy.

2.2 Why Simplify?

- Removing redundancy makes working with functional dependencies easier.
- It helps design efficient and clean databases.

2.3 Examples of Redundancy

Examples of Redundancy

- **Redundant FD:** In the set $\{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$, the FD $A \rightarrow C$ is redundant because it can be inferred from $A \rightarrow B$ and $B \rightarrow C$.
- **Redundant attributes on RHS:** $\{A \rightarrow B, B \rightarrow C, A \rightarrow CD\}$ can be simplified by removing C from $A \rightarrow CD$, giving $\{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$.
- **Redundant attributes on LHS:** $\{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ can be simplified by removing C from $AC \rightarrow D$, giving $\{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$.

2.4 Definition of Canonical Cover

Canonical Cover Properties

A set F_c of functional dependencies is a **canonical cover** for F if:

- F logically implies all dependencies in F_c .
- F_c logically implies all dependencies in F .
- No FD in F_c contains any extraneous attributes.
- Each left side of FD in F_c is unique.

2.5 How to Compute a Canonical Cover

How to Compute a Canonical Cover

To find a **canonical cover** for a set of functional dependencies F , follow these steps:

1. **Decompose each FD so that the right side has only one attribute:**

For any FD $\alpha \rightarrow \beta_1\beta_2 \dots \beta_n$, replace it with multiple FDs:

$$\alpha \rightarrow \beta_1, \quad \alpha \rightarrow \beta_2, \quad \dots, \quad \alpha \rightarrow \beta_n$$

This simplifies further analysis.

2. **Combine dependencies with the same left side (union rule):**

If you have multiple dependencies like $\alpha \rightarrow \beta_1$ and $\alpha \rightarrow \beta_2$, merge them into one:

$$\alpha \rightarrow \beta_1\beta_2$$

This reduces duplicates and simplifies the set.

3. **Remove unnecessary (extraneous) attributes:**

For each dependency $\alpha \rightarrow \beta$, check if any attribute in α (left side) or β (right side) can be removed without changing the meaning. Use attribute closure tests to decide if an attribute is extraneous and can be safely removed.

4. **Repeat these steps** until no further changes are possible.

Note: After removing extraneous attributes, re-check the union rule — new merges may be possible.

3 Example

Example 1

Example of Computing Canonical Cover

Consider relation:

$$R = (A, B, C)$$

with functional dependencies:

$$F = \{A \rightarrow BC, \quad B \rightarrow C, \quad A \rightarrow B, \quad AB \rightarrow C\}$$

Steps:

- Combine $A \rightarrow BC$ and $A \rightarrow B$ using union rule:

$$A \rightarrow BC$$

The set becomes:

$$\{A \rightarrow BC, \quad B \rightarrow C, \quad AB \rightarrow C\}$$

- Check if A is extraneous in $AB \rightarrow C$: Remove A and test if $B \rightarrow C$ holds — it does. So, remove $AB \rightarrow C$.
- Check if C is extraneous in $A \rightarrow BC$: Test if $A \rightarrow C$ can be derived from $A \rightarrow B$ and $B \rightarrow C$ — it can (transitivity). So, split into:

$$A \rightarrow B \quad \text{and} \quad A \rightarrow C$$

Then remove the extraneous C , keeping only:

$$A \rightarrow B$$

- Final canonical cover:

$$\{A \rightarrow B, \quad B \rightarrow C\}$$

Example 2: Computing the Canonical Cover

Example: Canonical Cover for Given FDs

Given:

$$F = \{A \rightarrow BC, \quad CD \rightarrow E, \quad B \rightarrow D, \quad E \rightarrow A\}$$

Find the canonical cover F_c .

Solution:

- Decompose FDs with multiple attributes on the right side:

$$A \rightarrow B, \quad A \rightarrow C, \quad CD \rightarrow E, \quad B \rightarrow D, \quad E \rightarrow A$$

- Check for extraneous attributes on left and right sides: none found.
- Check redundancy: none of the FDs can be derived from others alone.

Therefore, the canonical cover is:

$$F_c = \{A \rightarrow B, \quad A \rightarrow C, \quad CD \rightarrow E, \quad B \rightarrow D, \quad E \rightarrow A\}$$

Example 3

Example: Multiple Canonical Covers for a Set of FDs

Given the set of functional dependencies:

$$F = \{X \rightarrow YZ, \quad Y \rightarrow XZ, \quad Z \rightarrow XY\}$$

Step 1: Decompose FDs with multiple attributes on right side:

$$F_1 = \{X \rightarrow Y, \quad X \rightarrow Z, \quad Y \rightarrow X, \quad Y \rightarrow Z, \quad Z \rightarrow X, \quad Z \rightarrow Y\}$$

Step 2: Attempt to remove redundant dependencies or attributes:

- From F_1 , removing $X \rightarrow Z$ and keeping $Y \rightarrow Z$ and $Z \rightarrow Y$ still covers the original dependencies.

So one canonical cover could be:

$$F_a = \{X \rightarrow Y, \quad Y \rightarrow X, \quad Y \rightarrow Z, \quad Z \rightarrow X, \quad Z \rightarrow Y\}$$

- Alternatively, removing $Y \rightarrow Z$ and keeping $X \rightarrow Z$ and $Z \rightarrow Y$ also works, giving another canonical cover:

$$F_b = \{X \rightarrow Y, \quad X \rightarrow Z, \quad Y \rightarrow X, \quad Z \rightarrow X, \quad Z \rightarrow Y\}$$

Conclusion: Both F_a and F_b are minimal sets of dependencies logically equivalent to F , hence both are valid canonical covers.

This shows that a set of functional dependencies may have **more than one canonical cover**.

Example 4

Example: Risk of Removing Multiple Extraneous Attributes at Once

Using the same set of functional dependencies:

$$F = \{X \rightarrow YZ, \quad Y \rightarrow XZ, \quad Z \rightarrow XY\}$$

What can go wrong:

The canonical cover algorithm removes one extraneous attribute at a time. If multiple extraneous attributes are removed simultaneously, it may lead to loss of dependencies and incorrect results.

For example:

From $X \rightarrow YZ$, if both Y and Z are removed at once (assuming both extraneous), the FD reduces incorrectly to:

$$X \rightarrow \emptyset$$

which is meaningless and loses all information.

Correct approach:

Remove extraneous attributes *one at a time*, for example:

- Check if Y is extraneous in $X \rightarrow YZ$. If yes, remove it:

$$X \rightarrow Z$$

- Then check if Z is extraneous in $X \rightarrow Z$. Remove only if confirmed extraneous.

Conclusion: Removing extraneous attributes one at a time preserves the correctness of the canonical cover construction and avoids losing essential dependencies.